

Contrastive Multi-Knowledge Graph Learning for Fake News Detection

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Abstract—The widespread dissemination of fake news on social media platforms has significantly disrupted people’s daily lives, causing substantial inconvenience and loss. Although fake news detection has captivated great research effort, most of the existing works either overlook the complementary ground knowledge, formulated in individual knowledge graphs (KGs) or ignore multiple aspects of knowledge in different knowledge graphs. In this paper, we propose a novel contrastive learning framework for investigating the complementary knowledge in different knowledge graphs, which achieves superior fake news detection performance. We first model three components within news articles (i.e., news content, entities, and topics) as a heterogeneous graph (HG). Subsequently, we leverage the valuable factual evidence among real-world entities, which is enclosed in two open-source knowledge graphs (Wikidata and DBpedia), and augment the heterogeneous graph into three augmented graphs. We then fuse knowledge-enhanced high-level news representations from each augmented graph and extract a unified news representation through our proposed contrastive multi-knowledge graph learning framework, which is ultimately utilized to identify fake news. We conduct extensive experiments on four public fake news datasets and demonstrate the superiority of our method over nine state-of-the-art baselines regarding five widely-used evaluation metrics (i.e., accuracy, precision, recall, F1-score, and ROC).

Index Terms—Fake news detection, anomaly detection, knowledge graph, graph neural networks (GNNs), contrastive learning.

I. INTRODUCTION

THE popularity of social media, like Twitter, Instagram, Weibo, etc., has significantly stimulated and accelerated

the online intercommunication of people all over the globe. As is well-known, advanced communication techniques undoubtedly bring people closer together but come with the exponential growth of fake news produced on social media platforms. The rapid spread of fake news not only has an undesirable impact on individuals, such as damage to reputation, influence on decision-making, and erosion of trust but also causes serious consequences to society, like economic degradation, social division, and even public panic [1], [2]. Meanwhile, it is extremely challenging for individuals or organizations lacking prior knowledge to avoid being affected by fake news due to its complex content, underscoring the urgency of fake news detection [3], [4], [5].

To effectively identify fake news, recent efforts could be divided into two categories, i.e., natural language processing (NLP)-based methods and graph neural networks (GNNs)-based methods. The former methods investigate fake news detection as a text classification task, where NLP techniques (e.g., long short-term memory neural networks (LSTM), recurrent neural networks (RNNs) [6], convolutional neural networks (CNNs) [7], etc.) are applied to detect fake news from semantic features of news articles. However, approaches in this line fail to explore the news structural features, such as the relations among entities, topics, and news. Therefore, the latter methods model structural and textual information of news articles as graphs and employ GNNs for fake news detection [8]. The overall framework of GNN-based approaches can be summarized as: 1) First, the content and structure of news are extracted to construct informative graphs. 2) Second, the designed graph neural networks are used to generate efficient news representations. 3) Eventually, the news is classified as fake or real based on the representations. These works concentrate on modeling news articles and designing effective graph neural networks but ignore exploring the rich ground-truth knowledge in various knowledge graphs (KGs) for fake news detection.

The factual knowledge in KGs, such as Wikidata [9], DBpedia [10], and YAGO [11], has been widely used in the industrial community to determine investment opportunities, analyze the quality of products, etc. [12] The affluent knowledge in KGs can naturally serve as a factual basis for detecting fake news. However, previous works rarely focus on combining KGs and GNNs for fake news detection. Although CompareNet [13] attempts to use external knowledge for a simple extension of news embeddings, its straightforward approach cannot comprehensively explore KGs and leads to compromised performance.

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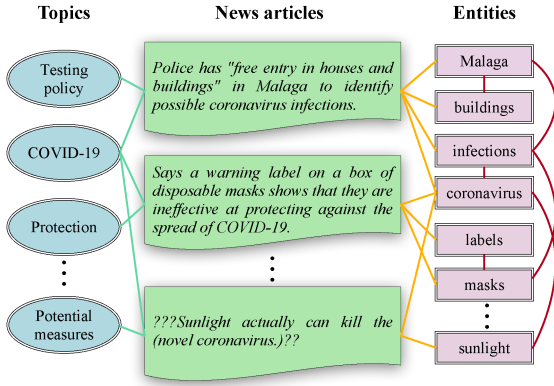


Fig. 1. The Construction of HG . HG is constructed in light of entities and related topics extracted from news content.

Additionally, the complementary role among multiple KGs has not been probed. Fake news on social media frequently spans various fields, making it challenging to rely solely on a single KG for comprehensive factual evidence in fake news detection. To be concrete, different KGs have their knowledge tendencies, which may limit their ability to address the diverse range of fake news circulating online. For example, a certain KG is abundant in medical knowledge, while another excels in financial knowledge [14], [15]. A single KG is insufficient for detecting fake news, as social media news often spans multiple domains. Thus, complementary knowledge from different KGs offers more comprehensive factual information for detecting fake news.

Existing studies (e.g., CompareNet [13]) combining a single KG with GNNs for fake news detection overlook the value of complementary knowledge from multiple KGs, limiting performance improvements. Therefore, we propose a novel **Contrastive Multi-Knowledge Graph Learning for Fake News Detection (Congrat)**, to provide fine-grained and multi-faceted ground-truth knowledge for enhancing detection performance, where multi-knowledge is the complementary factual evidence from different KGs. To be specific, we first model news content and structure as a heterogeneous graph (HG) with three types of nodes (i.e., news, topics, and entities), as shown in Fig. 1. Our approach thoroughly investigates the relationships among these nodes to obtain high-level news representations. To capture factual evidence for fake news detection, we then utilize the entities in two KGs to incorporate various factual information from different KGs into multi-view heterogeneous graphs and thereby enhance the integral graph content through graph augmentations. Then, we employ graph contrastive learning to assimilate the information of the constructed HG and two ground-truth KGs for learning news embeddings. The graph contrastive learning module possesses the capability to thoroughly absorb the complementary knowledge from multiple KGs, thereby offering substantial tangible evidence to detect fake news. After the learning process, we acquire valuable representations of news articles that encapsulate abundant information regarding both news content and KGs. These representations are subsequently utilized to accurately classify fake news from extensive news

data. Moreover, we conduct additional investigations to explore the impact of elements in the HG and KGs on our task. The main contributions of this research can be summarized as follows:

- To the best of our knowledge, we are the first to explore multiple aspects of knowledge encapsulated in different KGs for fake news detection. Factual knowledge in different KGs provides complementary information about the ground truth and is capable of serving as the factual basis for fake news detection.
- Our innovative Congrat framework introduces an adaptive capacity to digest ample textual and structural information from news articles. Simultaneously, it harnesses graph contrastive learning to effectively absorb complementary knowledge from two distinct KGs. Specifically, we incorporate complementary knowledge from two KGs to enhance our built HG through three different graph perspectives, as detailed in Fig. 2(①). Then, we employ graph contrastive learning to comprehensively compare factual information across these three graph perspectives. This allows us to investigate the impact of integrating knowledge from multiple KGs on fake news detection. Through extensive experiments, we shed light on the remarkable role of complementary knowledge from multiple KGs in improving the performance of fake news detection.
- We conduct comprehensive experiments¹ on four real-world datasets to verify the effectiveness of our proposed Congrat. The results demonstrate its superior performance compared with nine cutting-edge baselines on the evaluation metrics of accuracy, precision, recall, F1-score, and ROC. Simultaneously, the ablation study confirms the role of complementary knowledge from multiple KGs for enhancing fake news detection performance.

II. RELATED WORK

In this section, we summarize previous research in the following three fields: knowledge graph embedding, graph neural networks, and fake news detection.

A. Knowledge Graph Embedding

Knowledge graph embedding (KGE) target mapping triplets of KGs into a low-dimensional, dense, and interpretable space, thereby preserving as much knowledge as possible for downstream tasks like link prediction [16], anomaly detection [17], [18], [19], fake news detection [20], [21], [22], etc. KGE approaches can be categorized into three distinct groups: translation, semantic matching, and neural network models. The translation models attempted to interpret the relationships as a translational distance between head and tail entities, where the scoring function evaluates triplet plausibility by inversely correlating it with the magnitude of this distance [23], [24], [25]. The semantic matching models, e.g., RESCAL [26], QuatE [27], and ComplEx [28], aimed to qualify the semantic compatibility between entities and relations by employing scoring functions, which compute the similarity of their latent representations

¹<https://github.com/KG4FD/Congrat>

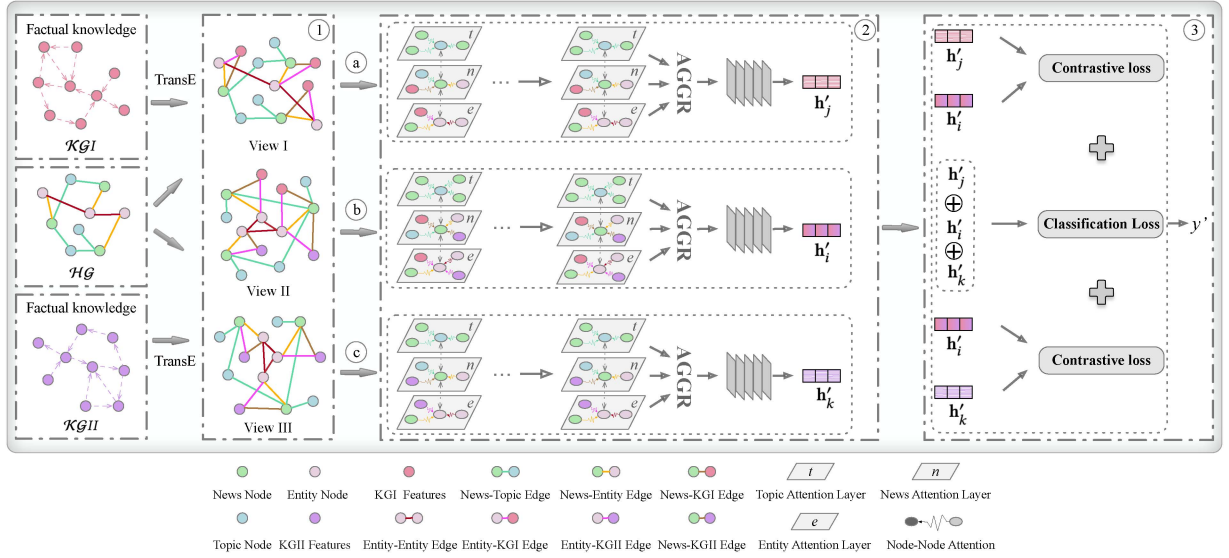


Fig. 2. The Framework of Congrat. Firstly, a heterogeneous graph H_G with three types of nodes (i.e. news, entities, and topics) is constructed according to news content. Then, two open-source knowledge graphs are leveraged to add ground-truth knowledge into the H_G for generating three augmented graph views (I, II, III), as shown in ②. Next, three views are input to attention-based GNNs for encoding news embeddings, detailed in 2. Afterward, the generated h'_j , h'_i and h'_k are used for learning the contrastive loss. Finally, we concatenate news embeddings by three views to predict news with label y' .

through dot product or tensor factorizations. The neural network models aspired to compute plausibility scores by jointly processing the head entities, relations, and tail entities through a neural architecture, which learns non-linear interactions to evaluate the semantic validity of KG triplets [29], [30], [31], [32].

B. Graph Neural Networks

In recent years, graph neural networks have emerged as a dominant paradigm for modeling topologically structured data, achieving exceptional performance by iteratively aggregating neighborhood information to capture structural and relational dependencies [33], [34], [35]. GNNs are compatible with data across numerous domains, which can be roughly categorized into graph convolutional networks, graph autoencoders, graph attention networks, graph generative networks, and graph spatial-temporal networks [36], [37], [38]. To date, a large number of scholars and organizations have attempted to use GNNs for various tasks (e.g., clustering [39], anomaly detection [40], multi-task learning [41], fake news detection [42], etc.), which achieved commendable results [43], [44]. In addition, the recent advent of graph contrastive learning (GCL) has significantly advanced the capabilities of GNNs. By extending the idea of contrastive learning, originally prevalent in visual representation tasks, to graph neural networks, GCL enables more generalizable, transferrable, and robust representation learning on graph structure data [45], [46], [47]. However, as an emerging methodology, GCL remains underexplored in diverse application contexts, particularly in the realm of fake news detection.

C. Fake News Detection

Recently, fake news detection techniques can be broadly divided into NLP-based and graph-based methods.

NLP-based approaches leverage traditional machine learning or deep learning techniques to analyze linguistic patterns and semantic cues from news articles' textual content, probably augmented by external knowledge bases, e.g., decision trees, support vector machines (SVM), RNNs, bidirectional LSTMs, CNNs, and other techniques [21], [48], [49], [50], [51]. Although these methods have a strong impression of enhancing detection performance, they do suffer from the drawback of neglecting news structures (e.g., the relations between news and topics [52], [53]). Such a limitation may constrain further advancements in detecting fake news. However, incorporating the structural elements of news articles into the detection workflow could be vital for achieving even higher levels of effectiveness in identifying fake news.

Therefore, graph-based approaches are introduced to consider the spatial structures of news articles in fake news detection. The methods of this line can be classified into social context-based and content-based approaches. The former aimed to represent the available auxiliary information on social networks (e.g., user comments, user profiles, user behaviors, etc. [54], [55], [56], [57]) as graphs to learn news embeddings for classifying whether the news is fake. However, accessing social media user information and profiles is usually challenging due to privacy concerns. In this research, we focus on content-based approaches, which targeted leveraging textual and structural features of news content (e.g., topics, entities, etc. [13]) to construct graphs for training GNNs to detect fake news. For example, Yao et al. [8] introduced TextGCN, a graph convolutional network that constructs a co-occurrence graph from textual features (e.g., word-document relationships) for text classification like fake news detection. Afterward, Karnyoto et al. [58] adapted TextGCN for fake news detection by integrating text augmentation, enhancing its robustness to linguistic variations in news content. In addition, HGAT and CompareNet, proposed by Hu et al. [13], [53], trained

GNNs on complex heterogeneous graphs to enhance detection capabilities, with CompareNet specifically incorporating KGs for the task. However, CompareNet simply concatenated news embeddings and KG embeddings, which resulted in the underutilization of the valuable information present in the KG knowledge.

Up to now, none of the former studies have integrated graphs and KGs into the learning process of GCL and investigated the impact of complementary knowledge from multiple KGs on fake news detection. Therefore, our work aims to bridge these research gaps by exploring news articles and diverse KGs with the specially designed Congrat framework.

III. PRELIMINARIES

A. Definition

1) *Definition 1: (Heterogeneous Graph):* A heterogeneous graph is defined as $\mathcal{HG} = \{\mathcal{N}, \mathcal{E}, \mathcal{A}, \mathcal{V}, \mathcal{R}\}$, where $\mathcal{N} = \{n_i\}_{i=1}^m$ denotes the set of nodes, $x_{ij} \in \mathcal{E}$ depicts an edge between node n_i and node n_j , $\mathcal{A} = \{a_i\}_{i=1}^m$ is the set of node attribute vectors, $\mathcal{V}$ and $\mathcal{R}$ represent the set of node types and edge types. A key condition for a heterogeneous graph is that $|\mathcal{V}| + |\mathcal{R}| > 2$, ensuring diversity in node and edge types.

2) *Definition 2: (Knowledge Graph):* A knowledge graph $\mathcal{KG}$ contains numerous triplets, each of which can be uniformly depicted as $\mathcal{KG} = \{ \langle s, r, t \rangle \}$, where s , t , and r are head entities, tail entities, and their relations, respectively.

B. Problem Statement

In this article, we define fake news as news articles on traditional and social media that contain factual inaccuracies or distortions. Therefore, graph-based fake news detection aims to detect anomalous/untrustworthy news with maximum accuracy performance. Given a heterogeneous graph $\mathcal{HG}$, each node n_i is a piece of news with an assigned label $y_i \in \{0, 1\}$, where 1 represents that the news is fake. When detecting, only a subset of node labels is known. After the detection procedure, the unseen fake news can be detected. In this paper, we attempt to use a $\mathcal{HG}$ defined in Section III-A1 and multiple $\mathcal{KG}$ s depicted in Section III-A2 to detect fake news with high accuracy.

IV. METHODOLOGY

In this part, we introduce our proposed **Contrastive Multi-Knowledge Graph Learning for Fake News Detection (Congrat)**. Section IV-A summarizes the general procedure of how our proposed model, Congrat, works superiorly. Sections IV-B to IV-E delve into the intricate details of the constituent elements within Congrat, specifically tailored to achieve outstanding fake news detection. Section IV-F provides a detailed explanation of the algorithm complexity associated with Congrat.

A. The Overview of Framework

To enhance fake news detection, we incorporate multiple ingredients into Congrat, i.e., three types of nodes in our built $\mathcal{HG}$, and the complementary knowledge from two $\mathcal{KG}$ s. The

overview of Congrat is shown in Fig. 2, which comprises four modules, i.e., data modeling, knowledge preserving news embedding, multi-source knowledge enhancement via contrastive learning, and fake news detector. Initially, we construct a heterogeneous graph $\mathcal{HG}$ for news articles and then use graph augmentation techniques, as shown in Fig. 2(①) to enhance the factual information in the $\mathcal{HG}$ with two $\mathcal{KG}$ s, Wikidata5M [59] and DBpedia [10]. Next, we input the knowledge-augmented graphs into the three designed graph attention networks (GAT) encoders to learn aggregated news embeddings by multi-edges in the $\mathcal{HG}$, detailed in Fig. 2(②). Afterward, we employ contrastive learning to compute the pairwise loss of the three knowledge-augmented graphs and subsequently use cross-entropy to calculate the overall classification loss, as shown in Fig. 2(③). In this manner, Congrat can detect fake news with superior performance.

B. Data Modelling

1) *Heterogeneous Graph Construction:* Social media platforms categorize the vast volume of published news articles based on their respective topics for efficient management. Hence, topics play a pivotal role as one of the primary properties defining real-world news. Furthermore, it is widely recognized that news articles typically encompass various entities that incorporate compelling and trending information to captivate the readers' interests. Thus, entities within news articles represent another essential property that contributes to the definition and understanding of news. Since topics and entities play significant roles in expressing the content of news articles, these two properties also offer valuable insights for fake news detection.

In this research, we build a $\mathcal{HG}$ with three types of nodes (i.e., news, entities, and topics) according to previous work [53], where $\mathcal{V} = \{n, t, e\}$ and $\mathcal{R} = \{x_{n,t}, x_{n,e}, x_{e,e}\}$. n , t and e denote news nodes, topic nodes, and entity nodes in the $\mathcal{HG}$ respectively. $x_{n,t}$, $x_{n,e}$ and $x_{e,e}$ represent news-topic edges, news-entity edges, and entity-entity edges, detailed in Fig. 1. We propose a series of steps designed to effectively construct the heterogeneous graph including node-level extraction, multi-edge connection, and feature initialization.

- *Node-Level Extraction:* In our built $\mathcal{HG}$, each news article is created as a news node with labels. For entity nodes, we leverage TAGME,² the entity link tool, to obtain various entities in news content. For topic nodes, we utilize the Latent Dirichlet Allocation (LDA) model to extract the topics of each news article. We set a threshold $\alpha = 50$ as the total number of topics for the LDA model and select the top K relevant topics (detailed in Section V-F) for each news article.
- *Multi-Edge Connection:* After creating nodes for $\mathcal{HG}$, the next step involves establishing connections for each pair of nodes. We build three types of edges in the $\mathcal{HG}$. The news-topic edge $x_{n,t}$ is connected when the news article is related to the topic. Each news node has K news-topic edges, detailed in Section V-F. The news-entity edge

²<https://sobigdata.d4science.org/group/tagme/>

$x_{n,e}$ is constructed when the news article contains the entity, while each news node potentially has numerous news-entity edges. The entity-entity $x_{e,e}$ is linked when the cosine similarity score surpasses a threshold $\beta = 0.5$.

— *Feature Initialization:* We employ natural language techniques (i.e., Doc2vec, term frequency-inverse document frequency (TF-IDF), and one-hot encoding) to obtain initial feature vectors for nodes in the $\mathcal{HG}$. For news nodes, TF-IDF and Doc2vec are leveraged to encode the content of each news article. In addition, TF-IDF and one-hot are applied to initialize the feature vectors for entity nodes and topic nodes. These methods are capable of capturing abundant features encapsulated in three types of $\mathcal{HG}$ nodes.

2) *Knowledge Graph Embeddings:* A knowledge graph $\mathcal{KG}$ encompasses valuable real-world knowledge that can provide factual information for fake news detection. Such knowledge is encapsulated in $\mathcal{KG}$ triplets consisting of head entities, relations, and tail entities. Nevertheless, it is noteworthy that the knowledge within a single $\mathcal{KG}$ might not cover all aspects of the world. Therefore, incorporating complementary knowledge from multiple $\mathcal{KG}$ s can provide more comprehensive and sufficient factual evidence for our task.

In this study, we employ $\mathcal{HG}$ entities to match relevant knowledge entities from two open-source $\mathcal{KG}$ s, ensuring that knowledge entities do not overlap with news nodes in $\mathcal{HG}$. Notably, we use Wikidata and DBpedia as factual knowledge for three reasons: 1) Both are founded on Wikipedia, a trusted and authoritative repository of factual knowledge. 2) Both are RDF-based $\mathcal{KG}$ s that can be readily queried with SPARQL [60]. 3) Both provide ample factual evidence to detect fake news, covering a broad spectrum of knowledge. 4) DBpedia extracts structured data from Wikipedia's infoboxes, whereas Wikidata offers secondary or tertiary data sourced from Wikipedia. Therefore, the knowledge within the two $\mathcal{KG}$ s strengthens each other in a mutually beneficial manner. After obtaining multiple $\mathcal{KG}$ knowledge entities, we expect to achieve excellent detection performance by exploring knowledge entity features. Various approaches that represent knowledge features in previous works (e.g., QuatE [27], TransE [23], etc.) have been proven effective in preserving valuable $\mathcal{KG}$ information while generating low-dimensional representations of knowledge entities, which can be efficiently utilized for downstream tasks. Our proposed framework, Congrat, can seamlessly integrate and collaborate with different $\mathcal{KG}$ embedding methods. Owing to the simplicity of TransE, we utilize the same embedding criterion and learn $\mathcal{KG}$ entity representations as the factual basis.

$$h_s + h_r \approx h_t \quad (1)$$

where s is the head entity, t represents the tail entity, and r denotes the relation between them. After training, each $\mathcal{KG}$ entity will be allocated a vector h_{kg} which is generated by moving s and the t closer to each other along the direction of r . we leverage h_{kg} of two $\mathcal{KG}$ s as factual evidence to implement heterogeneous graph augmentation.

3) *Graph Augmentation:* The techniques of graph augmentation can be generally split into feature-level, node-level, edge-level, and subgraph-level, graph-level approaches. In this study, we explore novel methods within the scope of node interpolation

and edge rewiring to augment our built $\mathcal{HG}$. This involves utilizing entities from two $\mathcal{KG}$ s to link news nodes and $\mathcal{HG}$ entity nodes, thereby altering the internal structure of the $\mathcal{HG}$ for augmentation purposes. Specifically, we incorporate factual background knowledge to expand the $\mathcal{HG}$ into three distinct augmentation views, detailed in Fig. 2(①). The augmentation operations can be summarized: 1) View I is enhanced by connecting the entities from $\mathcal{KGI}$ (i.e., Wikidata) with news nodes and entity nodes in the $\mathcal{HG}$, i.e., $x_{n,kgi}$ and $x_{e,kgi}$. 2) View II serves as the anchor of subsequent contrastive learning, which requires background knowledge from both $\mathcal{KG}$ s. Hence, it is augmented by linking the entities from two $\mathcal{KG}$ s with news and entities nodes in the built $\mathcal{HG}$, i.e., $x_{n,kgi}$, $x_{e,kgi}$, $x_{n,kgii}$ and $x_{e,kgii}$. 3) View III, similarly, is strengthened by building edges for the $\mathcal{KGII}$ (i.e., DBpedia) entities, news and entity nodes in the modeled heterogeneous graph, i.e., $x_{n,kgii}$ and $x_{e,kgii}$.

C. Knowledge Preserving News Embedding

After the acquisition of a content-based heterogeneous graph and two open-source knowledge graphs, features of the $\mathcal{HG}$ and $\mathcal{KG}$ s need to be encoded to acquire valuable news representations for promoting further fake news detection. In our proposed framework, we leverage the commonly used GAT to generate high-level news embeddings for the aforementioned augmented graph views. Specifically, we employ multiple attention schemes to learn news embeddings by aggregating neighboring information through the four-type relations surrounding news nodes in the three augmented graphs (i.e., $x_{n,e}$, $x_{n,t}$, $x_{n,kgi}$, $x_{n,kgii}$), detailed in Fig. 2(②). After getting news representations of different graph views, the contrastive module is introduced to distinguish positive samples from negative samples in high-dimensional spaces for training GNNs to generate the ultimate news embeddings for deep fake news detection.

1) *Node-Level Representation Generation:* There are three kinds of nodes in our built $\mathcal{HG}$ (i.e., news, entities, and topics). Congrat develops the graph attention encoder, as shown in Fig. 2(②), which targets generating high-level embeddings for nodes according to their relations and factual knowledge. For three types of nodes in $\mathcal{HG}$, the attention mechanism measures the impact of their neighbors based on the surrounding relations when projecting them into a latent space.

— *Graph View I Embedding Generation:* Particularly, for nodes in graphs, Congrat introduces distinct attention layers tailored to news, entities, and topics for aggregating neighboring information of nodes.

a) *Topic node embeddings:* Our approach leverages an attention neural network layer to generate their embeddings because the type of topic nodes' neighbors is only one (i.e., news nodes). Therefore, the attention layer follows news-topic relation $x_{n,t}$ to propagate node information via:

$$\mathbf{h}_{t,i}^{x_{n,t}} = a_{i,i}^{x_{n,t}} \mathbf{h}_{t,i}^{x_{n,t}} + \sum_{j \in \mathcal{S}^{x_{n,t}}(i)} a_{i,j}^{x_{n,t}} \mathbf{h}_{t,j}^{x_{n,t}}, \quad (2)$$

where $\mathbf{h}_{t,i}^{x_{n,t}}$ is the learned embedding of topic node i according to news-topic relation $x_{n,t}$. $\mathbf{h}_{t,i}^{x_{n,t}}$ and $\mathbf{h}_{t,j}^{x_{n,t}}$ represent the input feature vectors of topic i and its neighboring news node j respectively. $\mathcal{S}^{x_{n,t}}(i)$ denotes a set of surrounding news nodes

of topic i . $a_{i,i}^{x_{n,t}}$ and $a_{i,j}^{x_{n,t}}$ depict the self-attention parameter of topic i and the attention coefficient between topic i and news node j respectively, which can be calculated as follows:

$$a_{i,j}^\tau = \frac{\exp\left(\vec{\mathbf{b}}^\tau \left(\sigma[\mathbf{h}_{node,i}^\tau \parallel \mathbf{h}_{node,j}^\tau]\right)\right)}{\sum_{k \in S^\tau(i) \cup \{i\}} \exp\left(\vec{\mathbf{b}}^\tau \left(\sigma[\mathbf{h}_{node,i}^\tau \parallel \mathbf{h}_{node,k}^\tau]\right)\right)}, \quad (3)$$

where $\sigma(\cdot)$ represents an activation function like Tanh, LeakyReLU, etc. $\mathbf{h}_{node,i}^\tau$ and $\mathbf{h}_{node,j}^\tau$ express the feature representations of node i and its adjacent node j according to a specific relation τ . $\vec{\mathbf{b}}$ denotes trainable variables and $\parallel$ represents the “concatenate” operation for multiple matrices.

b) *Entity node embeddings*: Congrat designs three attention layers to digest the neighboring information for entity nodes according to three relations. According to a specific relation τ , the attention layer learns entity embeddings by:

$$\mathbf{h}_{e,i}^\tau = a_{i,i}^\tau \mathbf{h}_{e,i}^\tau + \sum_{j \in S^\tau(i)} a_{i,j}^\tau \mathbf{h}_{e,j}^\tau; \tau \in \{x_{e,e}, x_{n,e}, x_{e,kgi}\}, \quad (4)$$

where $\mathbf{h}_{e,i}^\tau$ represents the output representation of entity i according to a specific relation τ . $S^\tau(i)$ is a set of surrounding nodes of entity i according to the relation τ . $a_{i,i}^\tau$ and $a_{i,j}^\tau$ denote the self-attention of entity i and the attention between i and its adjacent nodes according to τ , which are quantized in the same way as (5).

c) *News node embeddings*: Congrat leverages three attention layers to generate news node embeddings according to the neighboring relations (i.e., news-topic, news-entity, news-kgi), and the method to obtain representations is measured as:

$$\begin{aligned} \mathbf{h}_{n,i}^\tau &= \frac{\exp\left(\vec{\mathbf{b}}^\tau \left(\sigma[\mathbf{h}_{n,i}^\tau \parallel \mathbf{h}_{n,i}^\tau]\right)\right)}{\sum_{k \in S^\tau(i) \cup \{i\}} \exp\left(\vec{\mathbf{b}}^\tau \left(\sigma[\mathbf{h}_{n,i}^\tau \parallel \mathbf{h}_{n,k}^\tau]\right)\right)} \mathbf{h}_{n,i}^\tau \\ &+ \sum_{j \in S^\tau(i)} \frac{\exp\left(\vec{\mathbf{b}}^\tau \left(\sigma[\mathbf{h}_{n,i}^\tau \parallel \mathbf{h}_{n,j}^\tau]\right)\right)}{\sum_{k \in S^\tau(i) \cup \{i\}} \exp\left(\vec{\mathbf{b}}^\tau \left(\sigma[\mathbf{h}_{n,i}^\tau \parallel \mathbf{h}_{n,k}^\tau]\right)\right)} \mathbf{h}_{n,j}^\tau, \end{aligned} \quad (5)$$

where $\mathbf{h}_{n,i}^\tau$ is the learned representations of news nodes, $S^\tau(i)$ is a set of neighboring nodes of news node i under a specific relation τ . Notably, $\tau \in \{x_{n,e}, x_{n,t}, x_{n,kgi}\}$.

- *Graph View II Embedding Generation*: Congrat generates the representations of topic nodes, and entity nodes according to (2) and (4), whereas the entity relations will be $\tau \in \{x_{e,e}, x_{n,e}, x_{e,kgi}, x_{e,kgi}\}$. The news embeddings are generated by (5) but the relation set will be $\tau \in \{x_{n,e}, x_{n,t}, x_{n,kgi}, x_{n,kgi}\}$.
- *Graph View III Embedding Generation*: Our model obtains topic embeddings and entity embeddings through (2) and (4), while the relation set of entity nodes will be $\tau \in \{x_{e,e}, x_{n,e}, x_{e,kgi}\}$. Similarly, the news representations are computed by (5) and the relations will be $\tau \in \{x_{n,e}, x_{n,t}, x_{n,kgi}\}$.

2) *News Representation Aggregation*: For the three augmented graph views, Congrat learns news embeddings through prior steps that comprise affluent and valuable information from

topics, entities, and two $\mathcal{KG}$ s. To generate unified news representations, we aggregate the news embeddings learned from the augmented graph views by the following equation:

$$\mathbf{h}'_i = \text{Aggr}(\mathbf{h}_{n,i}^\tau), \quad (6)$$

where $\text{Aggr}(\cdot)$ denotes an aggregation function. To avoid information loss, we utilize the “sum” operation to accumulate the obtained news representations $\mathbf{h}_{n,i}^\tau$ as $\mathbf{h}'_i$. For the three different graph views, τ will be distinguishable but belong to subsets of $\{x_{n,e}, x_{n,t}, x_{n,kgi}, x_{n,kgi}\}$.

D. Multi-Source Knowledge Enhancement Via Contrastive Learning

In this section, we introduce a graph contrastive loss to improve the performance of fake news detection. Generally, the news embeddings generated by GAT encoders only explore the internal information propagation in each graph view but fail to investigate the effect of complementary knowledge from multiple $\mathcal{KG}$ s on our task. To achieve this target, graph contrastive learning is a potent approach that can mine external complementary information between varying augmented graph views. Therefore, we design a contrastive module to further explore how complementary knowledge from $\mathcal{KG}$ s influences fake news detection.

Specifically, we generate three graph views to augment factual information for the original $\mathcal{HG}$ by capturing ground-truth knowledge from two open-source $\mathcal{KG}$ s. Then, we acquire news representations of each graph view through our devised GAT encoders. Next, we employ contrastive learning to enforce the embeddings of news nodes representing the same news article in the three graph views to agree with each other and be different from other news nodes. For any news node i , its embedding in view II, $\mathbf{h}'_i$ is treated as the anchor. The two embeddings of this news node i generated in the other graph views, $\mathbf{h}'_j$ and $\mathbf{h}'_k$ are treated as positive samples. The embeddings $\mathbf{h}'_q$ of the news nodes other than i in the three graph views are regarded as negative samples. Formally, we compute the contrastive loss of positive pairs by:

$$\mathcal{L}_{contras} = - \sum_{i \in \Phi} \log \frac{\exp(\theta(\mathbf{h}'_i, \mathbf{h}'_j)/\Omega) + \exp(\theta(\mathbf{h}'_i, \mathbf{h}'_k)/\Omega)}{\sum_{q \in \Delta(i)} \exp(\theta(\mathbf{h}'_i, \mathbf{h}'_q)/\Omega)}, \quad (7)$$

where θ is the cosine function to compute the similarity of two nodes. Ω denotes a temperature parameter. $\Phi = \{1, 2, \dots, 3m\}$ represents all news nodes generated by three graph views, each of which is m . $\Delta(i) = \Phi \setminus i$ denotes all negative samples. Since View I and View III are employed to commence contrastive learning with View II, we have two positive samples and $3m - 3$ negative samples for each news node. In addition, we can comprehensively investigate the impact of complementary knowledge from two $\mathcal{KG}$ s on the final detection performance by comparing three graph views.

E. Fake News Detector

The three graph attention encoders generate high-quality news embeddings by affecting each other via contrastive learning. For

fake news detection, Congrat concatenates each news representation learned by three graph views as the final news embeddings $\mathbf{h}'_{i,final}$ and feeds it to a multi-layer perceptron (MLP) for learning a deep fake news detector. The detector predicts the label of news i via:

$$y'_i = \Gamma(\rho(\mathbf{h}'_{i,final}; \pi)), \quad (8)$$

where $\rho(\cdot)$ denotes an activation function that outputs a two-dimensional vector with probabilities of news i 's label. $\Gamma(\cdot)$ selects the label with the maximum probability to assign to news i . π depicts trainable variables in MLP layers. The fake news classification loss function is defined as cross-entropy. Therefore, the overall loss of Congrat is measured as follows:

$$\mathcal{L}_{overall} = \lambda \mathcal{L}_{contras} + CrossEntropy(y_i, y'_i), \quad (9)$$

where y_i is the actual label of news i and y'_i is the predicted label. λ is the hyperparameter to control the contribution of the contrastive module. By minimizing the overall loss $\mathcal{L}_{overall}$, Congrat will predict news labels with high accuracy.

Notably, Congrat outputs the representations of all news nodes in $\mathcal{HG}$ and trains the detector by labels y_{train} of $\mathbf{N}_{train}$. When new data is input, our model extracts relevant entities and topics, integrating them into the existing graph for learning. Since the model's weights are not dependent on the size of $\mathcal{HG}$, we incrementally train the model with incoming data from the previous checkpoints, like training settings of large language models, demonstrating the real-world usability and adaptability of our model.

F. Algorithm Complexity

Algorithm 1 shows the details of Congrat. Apart from applying TransE for extracting knowledge entities' features, the multiple GAT layers and contrastive learning module introduce the most computational costs to Congrat. Specifically, the complexity of each GAT layer employed for generating news embeddings is approximately $\mathcal{O}(mDD' + |\mathcal{E}|D')$, where m is the total number of nodes in our constructed $\mathcal{HG}$, $|\mathcal{E}|$ is the total number of edges, $|D|$ is the input dimension and D' is the embedding dimension. The contrastive learning algorithm complexity is $\mathcal{O}(2m^2D')$ and the overall computational cost of Congrat will be $\mathcal{O}(mDD' + |\mathcal{E}|D' + 2m^2D') \approx \mathcal{O}(|\mathcal{E}|D' + m^2D') \approx \mathcal{O}(m^2D')$ since $D < m$ and $|\mathcal{E}| \leq m^2$.

V. EXPERIMENTS

In this section, we perform extensive experiments to address the following research questions:

- RQ1: How does the performance of Congrat compare to that of other state-of-the-art (SOTA) baselines for fake news detection?
- RQ2: How does complementary knowledge involved in $\mathcal{KG}$ s help improve performance?
- RQ3: How does the contrastive learning (CL) module benefit overall achievement?
- RQ4: How does Congrat perform when using different hyperparameters?

Algorithm 1: Congrat Framework.

Input: News articles $\mathbf{N}$; Two knowledge graphs $\mathcal{KGI}$, $\mathcal{KGII}$; Training news $\mathbf{N}_{train}$ and news labels y_{train} ; Classifiers $\rho(\cdot)$; Training iterations I .

Output: predict labels y' .

- 1 Build news content-based heterogeneous graph $\mathcal{HG}$ using $\mathbf{N}$
- 2 Extract complementary ground-truth entity features from $\mathcal{KGI}$ and $\mathcal{KGII}$
- 3 Graph augmentation: generate View I , View II , and View III
- 4 Randomly initialize all attention vectors for each relation $\vec{\mathbf{b}}^\tau$, $\tau \in \{x_{n,e}, x_{n,t}, x_{e,e}, x_{n,kgi}, x_{n,kgii}, x_{e,kgi}, x_{e,kgii}\}$
- 5 Randomly initialize π in $\rho(\cdot)$
- 6 $i \leftarrow 1$
- 7 **while** $i \leq I$ **do**
- 8 **for** V in View I , View II , and View III **do**
- 9 Update topics' representation $\mathbf{h}'_{t,i} \leftarrow$ Eq. 2;
- 10 Update entities' representation $\mathbf{h}'_{e,i} \leftarrow$ Eq. 4;
- 11 Update news' representation $\mathbf{h}'_{n,i} \leftarrow$ Eq. 5;
- 12 Obtain unified news representation $\mathbf{h}'_i \leftarrow$ Eq. 6;
- 13 **end**
- 14 Calculate contrastive loss $\mathcal{L}_{contras} \leftarrow$ Eq. 7;
- 15 Concatenate news embeddings of three graph views $\mathbf{h}'_{i,final}$;
- 16 Obtain predicted labels $y'_i \leftarrow$ Eq. 8;
- 17 Calculate the overall loss $\mathcal{L}_{overall}$ on the training set by Eq. 9;
- 18 Update $\vec{\mathbf{b}}^\tau$ and ρ to minimize $\mathcal{L}_{overall}$;
- 19 **end**
- 20 **Return:** $y' = \{y'_i\}_{i=1}^{|\mathbf{N}|}$, $y'_i = 1$ if fake, otherwise 0.

TABLE I
STATISTICS OF THE DATASETS

Dataset	#News	#Entities	#Topics	#Fake
COVID-19	8945	6181	50	4893
FakeNewsNet	2351	13331	50	1366
PAN2020	361	981	28	171
Liar	4411	4849	50	2426

- RQ5: How does the number of topics for each news article influence the detection performance?
- RQ6: How computationally efficient is Congrat compared with baselines?

A. Experimental Setup

1) *Datasets:* To verify the performance of our proposed model, we utilize four widely used datasets to conduct experiments consisting of COVID-19, FakeNewsNet, PAN2020, and Liar. COVID-19 [61] contains news articles related to the 2019 coronavirus pandemic collected from diverse social media platforms, e.g., Twitter, Facebook, etc. FakeNewsNet [62] comprises news content and associated posts, encompassing political news and entertainment news that have been fact-checked by reputable sources such as Politifact and Gossipcop. PAN2020 [63] is a fake news dataset labeled by authentic platforms such as PolitiFact or Snopes. Liar [64] is a fake news dataset that collects concise statements with labels from PolitiFact. The statistical information of the datasets we used in this research is shown in Table I.

TABLE II
COMPARISON OF CONGRAT AND BASELINES

Method	KG Usage	KG Strategy	Methodology	Category
RNN	N/A	N/A	RNNs	NLP
BiLSTM	N/A	N/A	LSTMs	NLP
TextCNN	N/A	N/A	CNNs	NLP
Electra	N/A	N/A	Transformer	NLP
TextGCN	N/A	N/A	GCNs	Graph
AugTextGCN	N/A	N/A	GCNs	Graph
HGAT	N/A	N/A	GATs	Graph
KGInformedFD	Single	Concatenate	Transformer + KG	NLP
ComparNet	Single	Concatenate	GATs + KG	Graph
Congrat	Multiple	Model learning	GCL + Multi-KGs	Graph

2) *Baselines*: We conduct comparative experiments in comparison to nine SOTA baseline methods to demonstrate the superiority of our proposed Congrat. These baselines can be divided into text-based approaches, graph-based approaches, and $\mathcal{KG}$ -based approaches, and their detailed information is discussed as follows:

1) *Text-based approaches*

- *BiLSTM* [65]: The model utilizes bi-directional neural networks to represent the textual content of news and then detects fake news via obtained news representations.
- *RNN* [6]: The method identifies fake news from the real by employing recurrent neural networks to embed news articles.
- *TextCNN* [66]: The approach embeds news content by applying a 1D-convolutional network for fake news detection.
- *Electra* [67]: The technique employs a large language model, ELECTRA, to represent news articles for detecting fake news.

2) *Graph-based approaches*

- *TextGCN* [8]: The model constructs a simple graph for news textual and semantic content and then employs graph convolutional neural networks (GCNs) to classify fake news.
- *AugTextGCN* [58]: The approach uses text augmentation techniques for news data and detects fake news by training GCN models.
- *HGAT* [53]: The method models short texts as a heterogeneous graph and then utilizes their designed GCNs to classify these short texts.

3) *$\mathcal{KG}$ -based approaches*

- *KGInformedFD* [21]: The approach applies language models to embed news articles and simply combines them with entity embeddings from a $\mathcal{KG}$ for fake news detection.
- *ComparNet* [13]: The technique generates news embeddings by concatenating $\mathcal{KG}$ entity representations and news representations learned by GCNs. Next, it utilizes the generated news embeddings for fake news detection.

Table II shows the differences between our proposed approach and the baselines discussed in this article. Notably, there are some semi-supervised methods like positive-unlabeled learning [57] widely used in the detection of fake news to address the scarcity of labels. However, our work prioritizes a supervised framework for the integration of high-quality $\mathcal{KG}$ s and graph contrastive learning. Supervised training ensures precise

alignment of news content with $\mathcal{KG}$ entities, which is crucial for modeling complex entity-news dependencies. In contrast, semi-supervised approaches risk introducing noise or mislinked entities, compromising detection robustness. Furthermore, our contrastive learning framework relies on label-guided subgraph comparisons to distinguish fake/real news, requiring reliable supervision to avoid adversarial confusion. For fairness, we exclude comparisons with semi-supervised learning methods in our experiments.

3) *Experimental Settings and Implementation*: Our proposed Congrat framework employs a layered structure consisting of two sets of GAT layers aiming to acquire news high-quality news embeddings for different augmented graph views according to distinct types of relations accompanying news nodes, topic nodes, and entity nodes of the $\mathcal{HG}$. eLU and LeakyReLU are chosen as the activation functions for graph attention networks and contrastive learning. Meanwhile, Adam is used as the optimizer with the learning rate, $lr = 0.001$, and weight decay, $wc = 5e - 4$ when backpropagating messages and minimizing overall loss. We split each dataset into 80%, 10%, and 10% as a training set, a test set, and a validation set. The maximum of the training epoch is set as 300 with an early stop when $\mathcal{L}_{overall}$ converges. Unless explicitly specified, the parameter settings for all baselines are identical to those employed in our model. We use accuracy, precision, recall, F1-score, and ROC as evaluation metrics for experimental results and set the same hyper-parameters of metrics for all approaches. All experiments are carried out on the Linux server with 60G RAM and an NVIDIA Tesla V100 GPU.

B. Model Comparison (RQ1)

Overall Performance: We compare our model with nine SOTA baselines to detect fake news on four datasets. From the experimental results in Table III and Fig. 3, we make observations as follows:

- In terms of accuracy, precision, recall, and F1-score, Congrat demonstrates its superiority over the majority of baselines on COVID-19, FakeNewsNet, and PAN2020 (except for precision on COVID-19). On Liar, although BiLSTM and TextCNN achieve remarkable accuracy and precision, Congrat outperforms the baseline models in overall performance.
- Regarding the receiver operating characteristic (ROC) curve, Congrat demonstrates a significantly superior area under the ROC curve (AUC) compared to nine baselines on four datasets, as illustrated in Fig. 3.

Comparison with Text-based Approaches (G1): These four methods achieve decent performance on certain datasets (e.g., over 85% accuracy on COVID-19), yet they also exhibit subpar results (e.g., below 50% recall on Liar). Given the results, the baselines achieve commendable performance on COVID-19, particularly Electra. However, their results still lag behind that of Congrat on COVID-19 (approximately 1%–10%) across all metrics. On FakeNewsNet, all these approaches achieve acceptable accuracy and precision, whereas Electra’s performance is second only to Congrat across all metrics. On PAN2020, the

TABLE III
EXPERIMENTAL RESULTS (IN PERCENTAGE) OF CONGRAT IN COMPARISON WITH NINE EXCELLENT BASELINES ON FOUR DATASETS (MEAN $\pm$ STDV, BEST IN BOLD, SECOND BEST IN UNDERLINE)

Dataset	Metrics	G1				G2			G3		Ours Congrat
		RNN	BiLSTM	TextCNN	Electra	TextGCN	AugTextGCN	HGAT	KGInformedFD	CompareNet	
COVID-19	Accuracy	86.10 $\pm$ 0.31	87.74 $\pm$ 0.13	84.76 $\pm$ 0.09	88.32 $\pm$ 0.15	61.32 $\pm$ 2.03	61.82 $\pm$ 2.56	84.51 $\pm$ 0.05	80.99 $\pm$ 0.59	84.49 $\pm$ 0.10	89.08 $\pm$ 0.66
	Precision	85.67 $\pm$ 0.43	93.55 $\pm$ 0.11	88.17 $\pm$ 0.08	85.52 $\pm$ 0.42	50.38 $\pm$ 3.21	50.76 $\pm$ 3.53	84.06 $\pm$ 0.13	81.12 $\pm$ 0.70	84.18 $\pm$ 0.13	87.95 $\pm$ 1.19
	Recall	83.46 $\pm$ 0.36	78.33 $\pm$ 0.34	77.17 $\pm$ 0.20	88.18 $\pm$ 0.55	77.22 $\pm$ 3.88	77.58 $\pm$ 3.93	81.23 $\pm$ 0.14	80.80 $\pm$ 0.58	80.82 $\pm$ 0.19	88.39 $\pm$ 2.09
	F1-score	84.49 $\pm$ 0.32	85.21 $\pm$ 0.18	82.28 $\pm$ 0.10	87.02 $\pm$ 0.17	58.76 $\pm$ 3.18	59.17 $\pm$ 3.26	82.60 $\pm$ 0.05	80.87 $\pm$ 0.59	82.45 $\pm$ 0.12	88.15 $\pm$ 0.93
FakeNewsNet	Accuracy	59.39 $\pm$ 0.26	73.12 $\pm$ 0.22	70.49 $\pm$ 0.16	77.79 $\pm$ 0.25	73.34 $\pm$ 1.25	73.55 $\pm$ 1.63	75.51 $\pm$ 0.19	52.97 $\pm$ 1.73	76.55 $\pm$ 0.18	79.32 $\pm$ 1.73
	Precision	53.02 $\pm$ 3.81	69.00 $\pm$ 0.32	66.72 $\pm$ 0.11	70.18 $\pm$ 0.37	36.39 $\pm$ 4.46	36.71 $\pm$ 4.56	68.63 $\pm$ 0.21	53.14 $\pm$ 1.44	68.78 $\pm$ 0.22	72.32 $\pm$ 3.96
	Recall	4.67 $\pm$ 0.61	64.58 $\pm$ 0.34	67.36 $\pm$ 0.27	79.23 $\pm$ 3.35	30.64 $\pm$ 3.76	31.11 $\pm$ 3.93	75.71 $\pm$ 0.63	53.05 $\pm$ 1.47	79.01 $\pm$ 0.29	79.52 $\pm$ 7.07
	F1-score	7.87 $\pm$ 0.95	66.68 $\pm$ 0.29	67.04 $\pm$ 0.14	75.15 $\pm$ 1.65	33.23 $\pm$ 4.07	33.70 $\pm$ 4.13	71.81 $\pm$ 0.26	52.49 $\pm$ 1.95	73.50 $\pm$ 0.18	75.52 $\pm$ 3.15
PAN2020	Accuracy	52.47 $\pm$ 0.65	52.74 $\pm$ 0.39	50.82 $\pm$ 0.53	60.82 $\pm$ 2.06	55.13 $\pm$ 1.31	55.47 $\pm$ 1.75	63.70 $\pm$ 0.73	51.60 $\pm$ 3.78	63.56 $\pm$ 0.51	73.63 $\pm$ 3.04
	Precision	52.70 $\pm$ 0.52	55.41 $\pm$ 0.63	54.32 $\pm$ 0.64	56.81 $\pm$ 2.23	31.77 $\pm$ 3.90	32.21 $\pm$ 4.32	60.81 $\pm$ 0.90	53.18 $\pm$ 3.83	63.61 $\pm$ 0.70	74.57 $\pm$ 4.04
	Recall	76.86 $\pm$ 2.85	80.36 $\pm$ 2.86	76.80 $\pm$ 3.51	67.36 $\pm$ 2.92	26.25 $\pm$ 3.24	27.10 $\pm$ 3.70	72.35 $\pm$ 0.78	52.71 $\pm$ 3.16	76.31 $\pm$ 1.30	80.89 $\pm$ 5.97
	F1-score	59.82 $\pm$ 1.59	61.41 $\pm$ 1.31	57.74 $\pm$ 1.84	61.56 $\pm$ 1.15	28.62 $\pm$ 3.51	29.15 $\pm$ 3.57	65.82 $\pm$ 0.77	49.52 $\pm$ 4.37	68.28 $\pm$ 0.48	77.43 $\pm$ 2.68
Liar	Accuracy	54.59 $\pm$ 0.11	60.61 $\pm$ 0.15	58.77 $\pm$ 2.20	57.47 $\pm$ 0.40	54.97 $\pm$ 6.83	55.25 $\pm$ 6.86	56.10 $\pm$ 0.18	37.04 $\pm$ 0.79	54.95 $\pm$ 0.11	59.12 $\pm$ 1.88
	Precision	36.31 $\pm$ 2.09	59.47 $\pm$ 0.39	65.87 $\pm$ 3.05	57.77 $\pm$ 4.14	14.70 $\pm$ 24.3	14.90 $\pm$ 22.5	31.84 $\pm$ 1.52	51.36 $\pm$ 0.24	49.81 $\pm$ 0.17	54.66 $\pm$ 2.01
	Recall	1.24 $\pm$ 0.15	45.33 $\pm$ 0.58	21.20 $\pm$ 5.41	34.84 $\pm$ 13.2	21.19 $\pm$ 41.7	21.46 $\pm$ 40.5	50.10 $\pm$ 0.25	32.32 $\pm$ 0.73	50.33 $\pm$ 0.19	57.06 $\pm$ 5.28
	F1-score	2.34 $\pm$ 0.27	51.12 $\pm$ 0.31	51.06 $\pm$ 3.49	41.43 $\pm$ 11.2	13.84 $\pm$ 24.7	14.33 $\pm$ 25.6	50.71 $\pm$ 0.23	30.15 $\pm$ 0.79	50.03 $\pm$ 0.12	55.73 $\pm$ 2.82

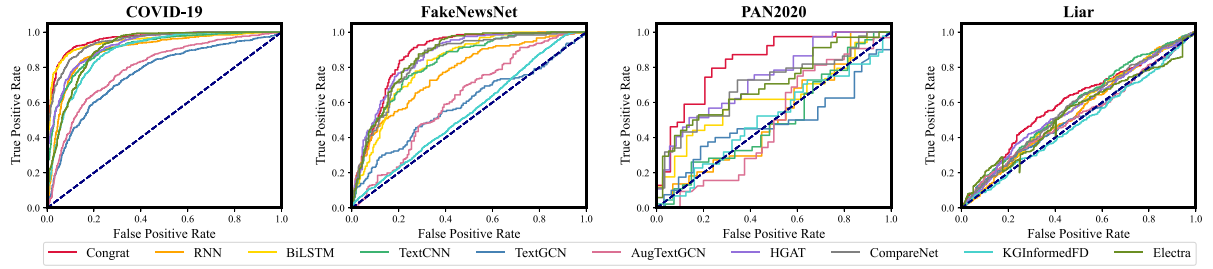


Fig. 3. The ROC Curves of Congrat and Baselines. To further verify the superiority of our model, the ROC curves on four datasets are provided to compare with the SOTA baselines, demonstrating Congrat's excellent performance.

four methods exhibit subpar performance except for the 80.36% recall of BiLSTM which ranks second only to Congrat. This is attributed to BiLSTM's ability to accurately predict true positive samples. On Liar, BiLSTM and TextCNN attain the best accuracy and precision but remain far below Congrat across other metrics. Besides, RNN performs worse than the other three models. In summary, Congrat significantly outperforms text-based methods, emerging as the leader in this category across most metrics.

Comparison with Graph-based Approaches (G2): These models do not exhibit notable performance on the four datasets regarding all metrics. Specifically, on COVID-19, TextGCN and AugTextGCN perform satisfactorily but lag behind Congrat, while HGAT performs well across all metrics but still falls short. On FakeNewsNet, the three methods achieve excellent accuracy, while other metrics are far lower than Congrat (except HGAT). On PAN2020, TextGCN and AugTextGCN exhibit subpar precision, recall, and F1-score despite attaining reasonable accuracy. However, HGAT achieves the second-highest accuracy (63.70%). On Liar, the models achieve acceptable accuracy but have unsatisfactory precision, recall, and F1-score. Notably, TextGCN and AugTextGCN perform unstably on different datasets. In addition, the significant variance in Liar's results is due to its inherent attributes, such as diverse topics and multiple sources, which may cause fluctuations in performance across different methods. Overall, our approach outperforms this group of methods significantly across all metrics.

Comparison with KG-based Approaches (G3): KGInformedFD and CompareNet leverage KGs to enhance fake news detection, improving their reliability in distinguishing fake news. On COVID-19, both approaches perform acceptably across all metrics, achieving scores above 80%. On FakeNewsNet, KGInformedFD underperforms (below 60% for all metrics), while CompareNet shows strong performance. On PAN2020, both approaches achieve acceptable results across four metrics, with CompareNet's precision and F1-score ranking second highest among all models. On Liar, KGInformedFD obtains subpar results while CompareNet performs commendably (nearly above 50% for all metrics), indicating that KGs can indeed enhance fake news detection. In conclusion, both methods exhibit decent performance on four datasets, while Congrat's detection capability still surpasses theirs.

Analysis of HG's Structure: From the experimental statistics, Congrat's performance varies across different datasets. As we can see, COVID-19 is the top-performing dataset, followed by FakeNewsNet, PAN2020, and Liar. The main reason is the coverage of entities in news nodes within KGs. Each news node is linked to at least three entity nodes, which can be associated with KG entities. Longer news texts contain more entities, strengthening their connection to KGs and providing more factual evidence. However, the shorter texts in the Liar dataset extract fewer entities, resulting in weaker associations with KGs.

Case Study: To qualitatively evaluate the detection results of our model, we provide four representative cases in Fig. 4,

Ground truth: True	Prediction: Fake	Type: FP
News 1: Sir Keir Starmer is out of isolation after his child tested negative for #coronavirus. However, he still won't be appearing at #PMQs today with Labour deputy leader Angela Rayner to face Boris Johnson in the House of Commons instead.		
Ground truth: Fake	Prediction: True	Type: FN
News 2: The Nazis were never democrats and never real socialists either." While there was a longstanding and distinguished Social Democratic Party in Germany from the 1870 to the 1920s, Lane wrote, the Nazis fought against it, and after 1933 imprisoned its leaders.....		
Ground truth: Fake	Prediction: Fake	Type: TP
News 3: Twin corps members celebrate their mum on birthday Photos of cute twin corps members and their mother has gone vir #URL# Photo: Mother & 5 years old daughter's bodies dumped under bridge, after mother refuses him sex. Police in Tha #URL# Alex Ekwueme is dead,		
Ground truth: True	Prediction: True	Type: TN
News 4: Says Donald Trump started his career back in 1973 being sued by the Justice Department for racial discrimination because he would not rent apartments in one of his developments to African-Americans.		

Fig. 4. Examples of specific news article detection results. The four possible outcomes (i.e., FP, FN, TP, and TN) of the fake news detection task are provided for visual illustration.

including false positive (FP), false negative (FN), true positive (TP), and true negative (TN). FP/FN cases often arise from politicized language, $\mathcal{KG}$ staleness, or misleading structural patterns in the contrastive graph. For example, during graph contrastive training, Congrat learns to distinguish ‘fake’ and ‘true’ by maximizing agreement between semantically similar nodes. However, structurally similar but semantically distinct cases (e.g., factual reports about political figures) may be pushed closer to the ‘fake’ cluster if $\mathcal{KG}$ s lack nuanced relational edges (e.g., ‘Sir Keir Starmer’ $\rightarrow$ ‘COVID-19 isolation guidelines’ $\rightarrow$ ‘public health protocols’). Fortunately, Congrat demonstrates strong detection performance across multiple datasets (i.e., TP and TN). This success can be attributed to the solid factual foundation provided by the complementary knowledge of $\mathcal{KG}$ s and the comprehensive training enabled by graph contrastive learning.

Generally speaking, our model exhibits apparent advantages in comparison to the other nine competing methods. The primary reasons for the superiority are as follows: 1) Congrat leverages different strategies to learn node embeddings in $\mathcal{HG}$ (e.g., LDA model, Doc2vec, etc.) and aggregate the information for the final generation of high-quality news representations according to multiple relations. 2) Meanwhile, GAT-based encoders are employed to adaptively learn the node representations of three augmented graphs, which can extract valuable information for the final news embeddings. 3) Congrat introduces contrastive learning to absorb complementary knowledge from different $\mathcal{KG}$ s to detect fake news. Multiple $\mathcal{KG}$ s are digested by our model, which can provide comprehensive factual knowledge to

TABLE IV
ABLATION TEST ON FOUR DATASETS (MEAN $\pm$ STDV, BEST IN BOLD)

Dataset	Variants	Accuracy	Precision	Recall	F1-score
COVID-19	Congrat	89.08$\pm$0.66	87.95 $\pm$ 1.19	88.39 $\pm$ 2.09	88.15$\pm$0.93
	Congrat-M1	87.53 $\pm$ 0.77	92.32 $\pm$ 5.56	79.34 $\pm$ 1.64	85.34 $\pm$ 1.06
	Congrat-M2	88.04 $\pm$ 0.63	86.52 $\pm$ 0.52	86.19 $\pm$ 0.73	87.52 $\pm$ 0.21
	Congrat-M3	85.80 $\pm$ 0.47	93.44$\pm$6.32	73.83 $\pm$ 7.36	82.48 $\pm$ 3.36
	Congrat-M4	85.75 $\pm$ 0.35	80.08 $\pm$ 0.72	91.86$\pm$5.30	85.57 $\pm$ 0.32
	Congrat-M5	80.60 $\pm$ 0.41	76.21 $\pm$ 0.46	83.42 $\pm$ 0.83	79.65 $\pm$ 0.56
	Congrat-M6	78.34 $\pm$ 7.36	92.20 $\pm$ 2.64	53.86 $\pm$ 20.7	66.36 $\pm$ 20.6
FakeNewsNet	Congrat	79.32$\pm$1.73	73.32$\pm$3.96	79.52$\pm$7.07	75.52$\pm$3.15
	Congrat-M1	77.28 $\pm$ 0.98	70.53 $\pm$ 0.52	76.04 $\pm$ 1.96	73.18 $\pm$ 0.37
	Congrat-M2	76.22 $\pm$ 1.23	69.80 $\pm$ 2.03	73.44 $\pm$ 3.64	71.57 $\pm$ 3.39
	Congrat-M3	76.86 $\pm$ 0.72	69.67 $\pm$ 1.33	76.56 $\pm$ 0.46	72.95 $\pm$ 0.83
	Congrat-M4	75.58 $\pm$ 1.23	66.96 $\pm$ 1.58	79.17 $\pm$ 0.63	72.55 $\pm$ 0.12
	Congrat-M5	71.55 $\pm$ 0.72	60.02 $\pm$ 1.26	79.12 $\pm$ 1.87	68.25 $\pm$ 1.45
	Congrat-M6	64.25 $\pm$ 7.01	46.58 $\pm$ 36.4	20.76 $\pm$ 26.9	27.80 $\pm$ 28.7
PAN2020	Congrat	73.63$\pm$3.04	74.57$\pm$4.04	80.89$\pm$5.97	77.43$\pm$2.68
	Congrat-M1	69.86 $\pm$ 6.18	66.67 $\pm$ 7.41	77.78 $\pm$ 12.9	71.79 $\pm$ 8.34
	Congrat-M2	68.49 $\pm$ 2.33	72.09 $\pm$ 3.77	73.81 $\pm$ 4.02	72.94 $\pm$ 5.66
	Congrat-M3	67.12 $\pm$ 2.55	69.44 $\pm$ 6.43	77.14 $\pm$ 3.61	69.23 $\pm$ 3.42
	Congrat-M4	64.38 $\pm$ 5.31	70.73 $\pm$ 3.94	67.44 $\pm$ 5.83	69.05 $\pm$ 4.60
	Congrat-M5	61.64 $\pm$ 7.56	63.64 $\pm$ 8.23	70.02 $\pm$ 5.32	66.67 $\pm$ 6.11
	Congrat-M6	56.46 $\pm$ 2.56	48.27 $\pm$ 3.78	73.58 $\pm$ 5.81	63.94 $\pm$ 1.86
Liar	Congrat	59.12$\pm$1.88	54.66$\pm$2.01	57.06$\pm$5.28	55.73$\pm$2.82
	Congrat-M1	56.74 $\pm$ 1.79	51.11 $\pm$ 1.93	53.48 $\pm$ 5.08	54.74 $\pm$ 2.87
	Congrat-M2	56.06 $\pm$ 3.36	53.13 $\pm$ 4.62	51.33 $\pm$ 2.85	52.22 $\pm$ 1.98
	Congrat-M3	56.06 $\pm$ 2.05	53.33 $\pm$ 3.26	50.72 $\pm$ 4.22	51.98 $\pm$ 5.37
	Congrat-M4	55.08 $\pm$ 2.76	50.26 $\pm$ 3.90	49.74 $\pm$ 2.87	50.00 $\pm$ 3.38
	Congrat-M5	54.93 $\pm$ 1.23	47.53 $\pm$ 2.55	32.44 $\pm$ 9.67	38.58 $\pm$ 8.34
	Congrat-M6	54.60 $\pm$ 0.00	00.00 $\pm$ 0.00	00.00 $\pm$ 0.00	00.00 $\pm$ 0.00

enrich news information for detecting fake news. Such knowledge provides factual evidence for news articles and the performance could improve with an increased amount of ground-truth knowledge provided.

C. Ablation Study (RQ2)

As shown in Table IV, we conduct an ablation study to explore how the performance of Congrat changes when different components (topics, entities, $\mathcal{KGI}$, $\mathcal{KGII}$, and the CL module) are removed. In the table, ‘Congrat-M6’ refers to a version where only the initial news embeddings without any element mentioned above are used for fake news detection. ‘Congrat-M5’ uses the representations of both topic nodes and news nodes to train GNNs for detecting fake news. ‘Congrat-M4’ only leverages the embeddings of entity, topic, and news nodes for detection. ‘Congrat-M3’, ‘Congrat-M2’, and ‘Congrat-M1’ successively incorporate $\mathcal{KGI}$, $\mathcal{KGII}$, and both $\mathcal{KG}$ s into the model, but without the CL module. Finally, ‘Congrat’ represents the full model with the CL module used to detect fake news.

According to ablation experiments, ‘Congrat-M6’ attains subpar performance (e.g., 27.80% precision on FakeNewsNet) except 92.20% precision on COVID-19 when only considering the initial news feature vectors. When topics are incorporated into our model, the performance of ‘Congrat-M5’ is significantly improved by approximately 2%–29% on COVID-19, 7%–50% on FakeNewsNet, 3%–15% on PAN2020 and 32%–48% on Liar, indicating that topics have the potential to enhance detection performance. When considering news, topics, and entities in our model for fake news detection, ‘Congrat-M4’ achieves high precision (79.17%) on FakeNewsNet and further improves performance on all four datasets. This implies that entities exert a substantial influence on the overall performance of the model. When we integrate knowledge graphs into our model, ‘Congrat-M3’, ‘Congrat-M2’, and ‘Congrat-M1’ perform better than the models without $\mathcal{KG}$ s (i.e., ‘Congrat-M6’, ‘Congrat-M5’)

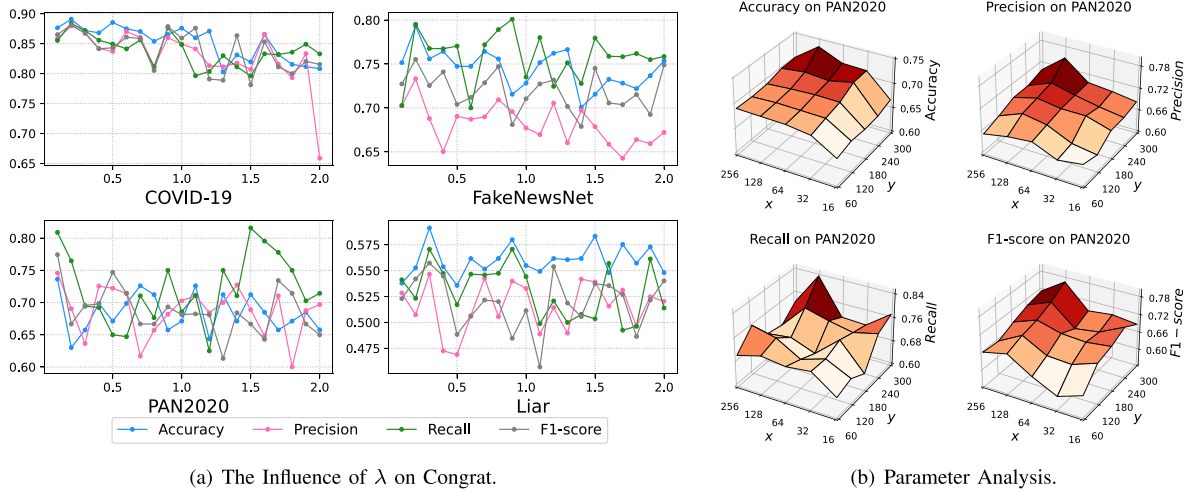


Fig. 5. Results of λ and Parameter Sensitivity. To explicitly compare the performance of our model, the hyperparameter λ test of models on four datasets is provided on the left side. To investigate parameter sensitivity, the 3D figure is presented on the right part, where the x-axis depicts embedding sizes and the y-axis denotes training epochs, detailed in Section V-E.

and ‘Congrat-M4’), which means that $\mathcal{KG}$ s indeed have a significant impact on fake news detection. Meanwhile, ‘Congrat-M3’ and ‘Congrat-M2’ exhibit similar performance across the four datasets, which indicates that mere substitution of a knowledge graph can not promote detection performance. When multiple $\mathcal{KG}$ s are considered to identify fake news, the model’s performance experiences a slight enhancement, indicating that only augmenting the quantity of knowledge graphs does not have a substantial influence on the model’s performance. When the contrastive learning module is added to our model, the complete model achieves the highest scores for almost all metrics on four datasets. This indicates that the CL module plays an important role in our model to digest knowledge from multiple $\mathcal{KG}$ s. The CL module facilitates the identification and absorption of complementary information across distinct $\mathcal{KG}$ s through comparison and then generates news representations encompassing comprehensive semantic, structural, and factual information.

In summary, news, topics, entities, $\mathcal{KGI}$, $\mathcal{KGII}$, and the CL module have a profound effect on the performance of fake news detection through ablation study. Notably, incorporating $\mathcal{KG}$ s into GNNs for training is significantly beneficial for performance improvement, and the CL module is extremely important to digest complementary knowledge from multiple $\mathcal{KG}$ s for fake news detection.

D. Contrastive Learning Module Analysis (RQ3)

In this part, we observe the impact of different λ on the overall classification performance. λ is the significant parameter to control the contribution of the contrastive loss to Congrat’s performance, detailed in Fig. 5(a). As shown in the figure, for different evaluation metrics, the values of λ present diverse trends across different datasets. On COVID-19, the performance of our model is optimal when $\lambda = 0.2$ in terms of the balance of all metrics. On FakeNewsNet, our model exhibits significant performance fluctuations according to different λ but Congrat

tends to be stable and optimal when $\lambda = 0.2$. On PAN2020, Congrat achieves the best performance when we set λ as 0.1. Obviously, $\lambda = 0.3$ is the optimal parameter for Liar, and our model showcases the highest level of discriminative power regarding accuracy, precision, recall, and F1-score. Based on the aforementioned observations, it is apparent that the value of λ must be adjusted according to the specific dataset. This adjustment allows for better control over the influence of the contrastive learning module on the overall model, thereby optimizing the model’s effectiveness.

E. Parameter Sensitivity (RQ4)

To assess the sensitivity of parameters, we conduct an evaluation of Congrat using various embedding sizes and training epochs in GNNs. For illustration purposes, we specifically examine the performance using PAN2020, considering 25 different combinations of epochs and embedding sizes. The embedding dimension and epoch are denoted as x and y , respectively, detailed in Fig. 5(b). The results demonstrate that the model’s performance exhibits improvement as the parameters increase, peaking at $x = 128$ and $y = 300$. Notably, our model consistently maintains satisfactory results across different parameter combinations.

F. Analysis of Topic Quantity (RQ5)

To observe the effect of the number of topics per news article on the model, we conduct experiments to analyze the fluctuating trend of Congrat’s performance under varying topic numbers, detailed in Fig. 6. In typical news articles, the number of topics generally remains below ten, otherwise, the news becomes overly scattered and devoid of coherence. Therefore, we set the number of topics per news article from 1 to 10. In the radar map, the outermost values denote the number of topics, while the radar scale represents the percentage corresponding to accuracy, precision, recall, and F1-score. As a metric extends outward, the

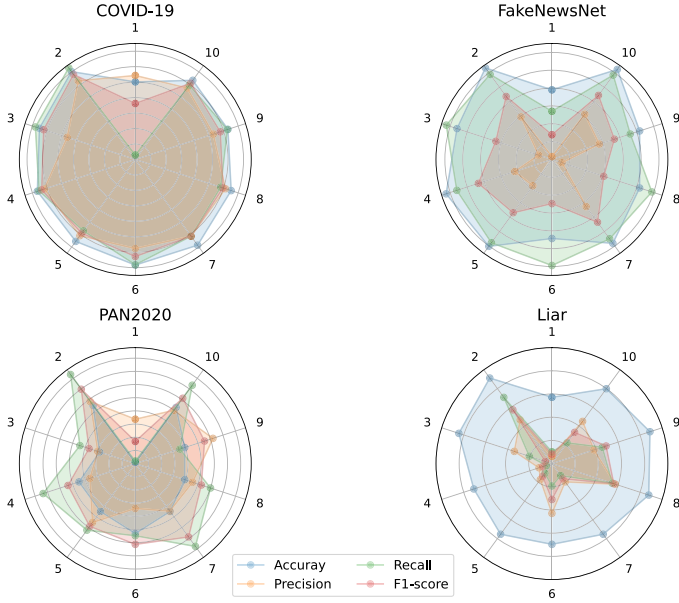


Fig. 6. Analysis of the number of topics per news article. To further demonstrate how the topic quantity influences our model, Radar maps on four datasets are provided to investigate Congrat’s performance when the number of topics per news article changes.

TABLE V
RUNNING TIME & GPU MEMORY COST

Method	COVID-19		FakeNewsNet	
	Time (s/epoch)	Mem (MB)	Time (s/epoch)	Mem (MB)
RNN	3.06	884.13	0.85	523.46
BiLSTM	3.79	836.79	1.07	502.82
TextCNN	1.22	909.63	0.37	639.23
Electra	0.20	2238.07	0.09	930.65
TextGCN	0.06	865.14	0.21	506.74
AugTextGCN	0.09	890.17	0.28	690.14
HGAT	0.65	1563.69	0.33	856.31
KGInformedFD	1.13	2507.15	0.67	1246.12
CompareNet	1.24	2679.91	0.88	1638.09
Congrat	0.68	2021.56	0.36	1034.56

performance of the metric improves. According to the results on four datasets, it is apparent that the model is in optimal condition with a balance of all metrics when the number of topics per news article is set as 2. Therefore, during the construction of the $\mathcal{HG}$, we set $K = 2$ as the top relevant topics of each news node.

G. Computational Cost (RQ6)

We compare the computational efficiency of Congrat with baselines in terms of both time and memory consumption, as detailed in Table V. We choose COVID-19 and FakeNewsNet as the representative datasets. From the table, it is evident that Congrat not only demonstrates an advantage in terms of computational cost but also delivers superior performance in a balanced manner. Specifically, Congrat achieves an optimal trade-off between moderate running time and memory usage, outperforming the baseline models that typically exhibit either higher computational costs or less efficient performance.

VI. CONCLUSION AND FUTURE WORKS

In this paper, we develop a novel contrastive multi-knowledge graph learning, namely Congrat to explore how complementary knowledge from different knowledge graphs influences fake news detection. To be concrete, we first construct a heterogeneous graph with three types of nodes (i.e., news, entities, and topics) based on news content. Then, we extract ground-truth knowledge entities from two different knowledge graphs to enhance the factual information for the $\mathcal{HG}$ into three augmented graph views. Next, we employ contrastive learning for the three graph views, which involves positioning positive samples in close proximity to each other within a high-dimensional space. Simultaneously, we ensure that positive and negative samples are placed far apart from each other within this space. Finally, we concatenate news embeddings learned from the three graph views and feed them to an MLP-based classifier for predicting news labels. Comprehensive experiments on real-world datasets are conducted to demonstrate the superiority of Congrat by comparing it with nine baselines, and the further ablation study verifies the efficacy of each part of our framework.

Discussion and Limitations: Our work focuses on fake news detection combining $\mathcal{KG}$ s and graph contrastive learning, with broader implications for combating fake news. The ‘truth sandwich’ strategy is to frame misinformation with accurate information before and after its presentation to mitigate false claims. Although our model doesn’t explicitly implement this structure, its use of $\mathcal{KG}$ -derived factual context aligns with the principle. By grounding predictions in verified entities and relationships (e.g., authoritative sources, temporal/logical consistency), our framework inherently prioritizes truth, similar to the ‘bread’ of a truth sandwich that surrounds misleading claims. We will explore integrating explicit ‘truth sandwich’ mechanisms into fake news detection in follow-up work. In addition, the computational complexity of integrating multiple knowledge graphs increases the resource requirements, which needs to find strategies for efficiency improvement.

Future Works: This work also identifies several under-explored areas for future research. These include exploring techniques to achieve improved performance when detecting short news articles within fake news datasets, determining which fake news topics should receive greater attention, such as those in the health or sports fields, and investigating advanced methods to capture more semantic features for detecting fake news in text-heavy news article data, etc. Furthermore, we will investigate alternative news data modeling approaches, such as weighted graphs, bi-directed graphs, and other explainable models.

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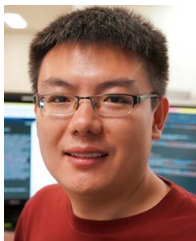
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